

SIDDHAANTH SATISH IYER

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EDUCATION

North Carolina State University | GPA: 3.5/4.0

M.S. Computer Engineering

Courses: ASIC Design with Verilog, Adv. Microprocessor Arch., ASIC Verification, Arch. of Parallel Computers, Hardware Security, ESL & PD

May 2027

Raleigh, NC

M.S Ramaiah Institute of Technology | GPA: 3.54/4.0

B.E. Electronics and Communication Engineering

Courses: Adv. Digital Design, Adv. Digital Logic Verification, Low Power VLSI Design

Oct 2024

Bengaluru, India

SKILLS

VLSI and ASIC Flow: RTL Design, Formal Verification, Design Verification (UVM), RTL-to-GDS / Physical Design, Synthesis, Place & Route, Static Timing Analysis, Computer Architecture, Low Power Design, ASIC Tape-out

Languages: SystemVerilog, Verilog, x86 Assembly, C/C++, Python, Bash, TCL, LaTeX

Tools: Cadence JasperGold, Synopsys Design Compiler, LibreLane / OpenROAD, AMD Xilinx Vivado, ModelSim, Magic / KLayout, MATLAB, Git / Linux

EXPERIENCE

GPU Formal Verification Intern

NVIDIA Corporation

Jan 2024 - Jul 2024

Bengaluru, India

- Closed formal verification of PCIe Transaction and Data Link Layer sub-units using SVA and Cadence JasperGold, achieving full property coverage for protocol compliance and corner-case scenarios.
- Collaborated with RTL designers to triage and resolve functional bugs discovered during FV, directly accelerating block sign-off.
- Optimized pre-submit FV automation scripts, cutting verification runtime by ~40% through targeted property pruning and constraint refinement.

Project Engineering Intern

Defense Research and Development Organization (DRDO) - DEBEL

Jul 2024 - Feb 2025

Bengaluru, India

- Designed a military-grade PCB for a mass-produced temperature-sensing wearable, integrating components under strict size, power, and environmental survivability constraints for the Indian Armed Forces.
- Developed custom ATmega328P and sensor firmware libraries with optimized interrupt handling, improving system response time by ~80%.

PROJECTS

Tape-out of a Minimum-Area Digital Baseband Transform Engine

Aug 2026

- Took a minimum-area OFDM / DFT-s-OFDM transform core from RTL to GDSII on the GlobalFoundries GF180MCU 180nm PDK using the open-source LibreLane flow (Yosys, OpenROAD, Magic, Netgen, KLayout) as an entry to the IEEE SSCS TC-OSE Chipathon.
- Folded a 12-point DFT and a 128-point FFT/IFFT onto a single shared mixed-radix datapath to minimize silicon area, hardening a 1.3 x 1.0 mm die at ~57% utilization with 2 SRAM macros and a clean LVS/XOR sign-off.
- Expecting a ~40% drop in power consumption when compared to existing architectures due to the folding logic.

Fixed-Point CORDIC/Kalman AHRS Coprocessor for Tape-out

Present

- Designing a merged-datapath attitude coprocessor for the TinyTapeout TTIHP26b shuttle (IHP SG13G2 130nm) that shares one shift-add iterator across CORDIC circular/linear modes and Kalman microcode, removing dedicated hardware multipliers and reciprocal ROMs to fit a 1x2 tile.
- Converts gyroscope/accelerometer/magnetometer streams into roll, pitch, and yaw in Q1.15 fixed point; authored a bit-exact Python golden model that emits the SV package and test vectors, and proved N_ITER=14 / W_INT=18 minimal via self-test.

Pipelined Convolution Neural Network Accelerator

Dec 2025

- Designed and synthesized a pipelined CNN accelerator in SystemVerilog for a 4x4 kernel over 1024x1024 inputs; achieved timing closure at a 7ns clock period via critical path optimization, passing all 6 functional test vectors with 100% correctness.
- Architected parallel MAC pipelines with ReLU, average pooling, and FIR filter stages across 35,393 standard cells in a 45nm library, sustaining 1 pixel/cycle throughput over 12.6M cycles of full-image processing.

Functional Verification of I2C to WishBone Multi-Bus Controller

Apr 2026

- Built a layered SystemVerilog/UVM testbench for an I2C-to-Wishbone bridge with driver, monitor, scoreboard, and SVA protocol checkers; verified all I2C transaction types (START/STOP, R/W, ACK/NACK) and Wishbone bus operations (cycle, strobe, acknowledge, data transfer).
- Structured the verification environment using UVM phases; collected functional coverage metrics achieving comprehensive coverage across transaction scenarios and corner-case conditions.

Differential Power Analysis Attack on AES-128

Sep 2026

- Implemented AES-128 encryption and decryption from scratch in C++ and verified against known-answer test vectors with 100% accuracy.
- Mounted a correlation power analysis attack in Python on a T-table AES RTL implementation, recovering all 16 bytes of round-key 10 from 10,000 simulated power traces and inverting the key schedule to recover the master key; exploited last-round byte-locality to reduce a 2^{128} search to sixteen independent 2^8 searches.

Computer Architecture Modeling Suite — OoO Superscalar, Caches, Branch Prediction

2025 - 2026

- Built cycle-accurate C++ models of an out-of-order superscalar pipeline (ROB, Issue Queue, RAT, Load/Store Queue) scaling IPC from ~1 to ~7.5 across WIDTH=1-8, and of Bimodal/GShare/Hybrid branch predictors where GShare (m=20, n=11) reached a 6.37% misprediction rate on gcc, a 43% reduction over the best Bimodal.
- Developed a parameterized L1/L2/L3 cache simulator with configurable associativity, LRU, and stream-buffer prefetching (99.6% L1 miss-rate reduction on streaming workloads), and modeled MSI/MESI/MOESI coherence for shared-memory multiprocessors.

PUBLICATIONS

Siddhaanth S Iyer, Dr. Deepali Koppad, Srirangarajan T M, “Wearable Temperature Detection System for Personalized Health Tracking”, *International Conference on Computational, Communication and Information Technology*, 2025.

Archita V, Ananya Kodukula, Siddhaanth S Iyer, “A Novel Approach to Mitigate Microgravity-Induced Bone Loss”, *International Astronautical Conference (IAC)*, 2022.

M S Dhanyavan, Siddhaanth S Iyer, Swagat Kumar Sarangi, “A Novel Degeneration and De-Orbiting Technique for Space Debris Removal”, *International Astronautical Conference (IAC)*, 2022.